

Savitribai Phule Pune University
T.Y.B.Sc.(C.S.) (NEP – 2020) SEM – V Practical Examination
Lab Course CS-305-MJ-P
Operating System-I

Duration: 3 Hours

Maximum Marks:35

Q.1) Write a C Menu driven Program to implement following functionality

- a) Accept Available
- b) Display Allocation, Max
- c) Display the contents of need matrix
- d) Display Available

Process	Allocation			Max			Available		
	A	B	C	A	B	C	A	B	C
P0	2	3	2	9	7	5	3	3	2
P1	4	0	0	5	2	2			
P2	5	0	4	1	0	4			
P3	4	3	3	4	4	4			
P4	2	2	4	6	5	5			

[15 Marks]

Q.2 Write a C program that behaves like a shell which displays the command

prompt '\$'. It accepts the command, tokenize the command line and execute it by creating the child process. Also implement the additional command 'count' as

- a. \$ count c filename: It will display the number of characters in given file
- b. \$ count w filename: It will display the number of words in given file
- c. \$ count l filename: It will display the number of lines in given file

[15 Marks]

Q.3. Oral/Viva

[5 Marks]

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Maximum Marks:35

Q.1 Write a program to simulate Linked file allocation method. Assume disk with 'n' number of blocks. Give value of 'n' as input. Randomly mark some block as allocated and accordingly maintain the list of free blocks Write menu driver program with menu options as mentioned below and implement each option.

- Show Bit Vector
- Create New File
- Show Directory
- Exit

[15 Marks]

Q.2 Write a simulation program for disk scheduling using FCFS algorithm. Accept total number of disk blocks, disk request string, and current head position from the user. Display the list of requests in the order in which it is served. Also display the total number of head moments.
55, 58, 39, 18, 90, 160, 150, 38, 184
Start Head Position: 50

[15 Marks]

Q.3. Oral/Viva

[5 Marks]

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Q.1 Write a C program to simulate Banker's algorithm for the purpose of deadlock avoidance. Consider the following snapshot of system, A, B, C and D is the resource type.

Process	Allocation				Max				Available			
	A	B	C	D	A	B	C	D	A	B	C	D
P0	0	0	1	2	0	0	1	2	1	5	2	0
P1	1	0	0	0	1	7	5	0				
P2	1	3	5	4	2	3	5	6				
P3	0	6	3	2	0	6	5	2				
P4	0	0	1	4	0	6	5	6				

- a) Calculate and display the content of need matrix?
b) Is the system in safe state? If display the safe sequence.

[15 Marks]

Q.2 Write a simulation program for disk scheduling using SSTF algorithm.
Accept total number of disk blocks, disk request string, and current head position from the user. Display the list of requests in the order in which it is served. Also display the total number of head moments.
98, 183, 37, 122, 14, 124, 65
Start Head Position: 53

[15 Marks]

Q.3. Oral/Viva [05 Marks]

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Q.1 Implement the Menu driven Banker's algorithm for accepting Allocation,
Max from user

- a) Accept Available
- b) Display Allocation, Max
- c) Find Need and display It,
- d) Display Available

Consider the system with 3 resources types A, B, and C with 7, 2, 6 instances respectively consider the following snapshot:

Process	Allocation			Request		
	A	B	C	A	B	C
P0	0	1	0	0	0	0
P1	4	0	0	5	2	2
P2	5	0	4	1	0	4
P3	4	3	3	4	4	4
P4	2	2	4	6	5	5

[15 Marks]

Q.2 Write a simulation program for disk scheduling using SCAN algorithm.

Accept total Number of disk blocks, disk request string, and current head position from the user. Display the list of requests in the order in which it is served. Also display the total number of head moments.
82, 170, 43, 140, 24, 16, 190, 65

Starting Head position= 50

Direction: Left

[15 Marks]

Q.3. Oral/Viva

[5 Marks]

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Maximum Marks:35

Q.1 Write the simulation program for demand paging and show the page scheduling total and number of page faults according the FIFO page replacement algorithm. Assume the memory of n frames.

Reference String: 3,4,5,6,3,4,7,3,4,5,6,7,2,4,6

[15 Marks]

Q2 Consider a system with 'n' processes and 'm' resource types. Accept number of instances for every resource type. For each process accept the allocation and maximum requirement matrices. Write a program to display the contents of need matrix and to check if the given request of a process can be granted immediately or not. (Use resource request algorithm)

[15 Marks]

Q.2. Oral/Viva

[5 Marks]

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Q.1. Write a program to simulate Sequential file allocation method. Assume disk with n number of blocks. Give value of n as input. Randomly mark some block as allocated and accordingly maintain the list of free blocks Write menu driver program with menu options as mentioned below and implement each option.

- Show Bit Vector
- Create New File
- Show Directory
- Exit

[15 Marks]

Q.2 Write a simulation program for disk scheduling using C-SCAN algorithm. Accept total number of disk blocks, disk request string, and current head position from the user. Display the list of requests in the order in which it is served. also display the total number of head movements (**Assume disk size=200**)
15, 45, 75, 105, 135, 165, 195, 80
Starting Head Position: 100
Direction: Right

[15 Marks]

Q.3. Oral/Viva

[05 Marks]

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Q.1 Consider the following snapshot of the system.

Process	Allocation				Max				Available			
	A	B	C	D	A	B	C	D	A	B	C	D
P0	2	0	0	1	4	2	1	2	3	3	2	1
P1	3	1	2	1	5	2	5	2				
P2	2	1	0	3	2	3	1	6				
P3	1	3	1	2	1	4	2	4				
P4	1	4	3	2	3	6	6	5				

Using Resource –Request algorithm to Check whether the current system is in
safe state or not [15 Marks]

Q2. Write a program that demonstrates the use of nice() system call. After a
child process is started using fork(), assign higher priority to the child using
nice()system call. [15 Marks]

Q.3. Oral/Viva [05 Marks]

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Q.1 Write a program to simulate Index file allocation method. Assume disk with 'n' number of blocks. Give value of 'n' as input. Randomly mark some block as allocated and accordingly maintain the list of free blocks Write menu driven program with menu options as mentioned above and implement each option

- Show Bit Vector
- Create New File
- Show Directory
- Exit

[15 Marks]

Q.2 Write a program that demonstrates the use of nice () system call. After a child process is started using fork(), assign higher priority to the child using nice() system call.

[15 Marks]

Q.3. Oral/Viva

[05 Marks]

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Q.1 Write the simulation program for demand paging and show the page scheduling and total number of page faults according the optimal page replacement algorithm. Assume the memory of 'n' frames.

Reference String: 8, 5, 7, 8, 5, 7, 2, 3, 7, 3, 5, 9, 4, 6, 2

[15 Marks]

Q.2 Write the program to simulate Non-preemptive Shortest Job First (SJF) scheduling. the arrival time and first CPU burst for different n number of processes should be input to the algorithm. The next CPU burst should be generated randomly. The output should give Gantt chart, turnaround time and waiting time for each process. Also find the average waiting time and turnaround time.

[15 marks]

Q.3. Oral/Viva

[5 Marks]

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Q.1 Write the program to simulate Round Robin (RR) scheduling. The arrival time and first CPU burst for different n number of processes should be input to the algorithm. Also give the time quantum as input. The next CPU burst should be generated randomly. The output should give Gantt chart, turnaround time and waiting time for each process. Also find the average waiting time and turnaround time. [15Marks]

Q.2 Write a simulation program for disk scheduling using Look algorithm. Accept total number of disk blocks, disk request string, and current head position from the user. Display the list of request in the order in which it is served. Also display the total number of head moments. 55, 58, 39, 18, 90, 160, 150, 38.
Start Head Position: 100
Direction: Left [15 Marks]

Q.3. Oral/Viva [5 marks]

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Maximum Marks:35

Q.1 Write a C program to simulate Banker's algorithm for the purpose of Deadlock avoidance. The following snapshot of system, A, B, C and D are the resource type.

Process	Allocation			Max			Available		
	A	B	C	A	B	C	A	B	C
P0	0	1	0	0	0	0	0	0	0
P1	2	0	0	2	0	2			
P2	3	0	3	0	0	0			
P3	2	1	1	1	0	0			
P4	0	0	2	0	0	2			

Implement the following Menu.

- a. Accept Available
- b. Display Allocation, Max
- c. Display the contents of need matrix
- d. Display Available

[15 Marks]

Q.2 Write the simulation program for demand paging and show the page scheduling and total number of page faults according the optimal page replacement algorithm. Assume the memory of n frames.

Reference String: 8, 5, 7, 8, 5, 7, 2, 3, 7, 3, 5, 9, 4, 6, 2

[15 Marks]

Q.3. Oral/Viva

[05 Marks]

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-
- Q.1 Write the program to simulate FCFS CPU scheduling. The arrival time and first CPU- burst for different n number of processes should be input to the algorithm. The next CPU-burst should be generated randomly. The output should give Gantt chart, turnaround time and waiting time for each process. Also find the average waiting time and turnaround time. [15 marks]
- Q.2 Implement the C program to accept n integers to be sorted. Main function creates child process using fork system call. Parent process sorts the integers using bubble sort and waits for child process using wait system call child process sorts the integers using insertion sort [15 marks]
- Q.3. Oral/Viva [5 Marks]

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Maximum Marks:35

-
- Q.1 Write the program to simulate Non-preemptive Shortest Job First (SJF) scheduling. The arrival time and first CPU burst for different n number of processes should be input to the algorithm. The next CPU-burst should be generated randomly. The output should give Gantt chart, turnaround time and waiting time for each process. Also find the average waiting time and turnaround time. [15 Marks]
- Q.2 Write a simulation program for disk scheduling using LOOK algorithm. Accept total number of disk blocks, disk request string, and current head position from the user. Display the list of requests in the order in which it is served. Also display the total number of head movements
15, 30, 55, 90, 125, 140, 170
Starting Head Position: 100
Direction: Left [15 Marks]
- Q.3. Oral/Viva [5 Marks]

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Maximum Marks:35

-
- Q.1. Implement the C program that accepts an integer array. Main function forks child process. Parent process sorts an integer array and passes the sorted array to child process through the command line arguments of `execve()` system call. The child process uses `execve()` system call to load new program that uses this sorted array for performing the binary search the particular item in the array.
[15 Marks]
- Q.2 Write the simulation program for demand paging and show the page scheduling and total number of page faults according the FIFO page replacement algorithm. Assume the memory of n frames.
Reference String: 3, 4, 5, 6, 3, 4, 7, 3, 4, 5, 6, 7, 2, 4, 6
[15 Marks]
- Q.3. Oral/Viva
[05 Marks]

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- Q.1 Write a C program to illustrate the concept of orphan process. Parent process creates a child and terminates before child has finished its task. So child process becomes orphan process. (Use fork(), sleep(), getpid(), getppid) [15 Marks]
- Q.2 Write the program to simulate Preemptive Priority scheduling. The arrival time and first CPU-burst and priority for different n number of processes should be input to the algorithm. The next CPU-burst should be generated randomly. The output should give Gantt chart, turnaround time and waiting time for each process. Also find the average waiting time and turnaround time . [15 Marks]
- Q.3. Oral/Viva [5 Marks]

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Q.1 Write the simulation program to implement demand paging and show the page scheduling and total number of page faults according to the LRU (using counter method) page replacement algorithm. Assume the memory of n frames.

Reference String : 3,5,7,2,5,1,2,3,1,3,5,3,1,6,2

[15 Marks]

Q.2 Write a C program that behaves like a shell which displays the command prompt '\$'. It accepts the command, tokenize the command line and execute it by creating the child process. Also implement the additional command 'count' as

- a \$ count c filename: It will display the number of characters in given file
- b \$ count w filename: It will display the number of words in given file
- c \$ count l filename: It will display the number of lines in given file

[15 Marks]

Q.3. Oral/Viva

[05 Marks]

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Q.1 Write the program to simulate Non-preemptive Shortest Job First (SJF) scheduling. The arrival time and first CPU-burst for different n number of processes should be input to the algorithm. The next CPU-burst should be generated randomly. The output should give Gantt chart, turnaround time and waiting time for each process. Also find the average waiting time and turnaround time [15 Marks]

Q.2 Write a program to simulate Sequential (Contiguous) file allocation method. Assume disk with n number of blocks. Give value of n as input. Randomly mark some block as allocated and accordingly maintain the list of free blocks. Write menu driven program with menu options as mentioned below and implement each option.

- a. Show Bit Vector
- b. Create New File
- c. Show Directory
- d. Delete File
- e. Exit

[15 Marks]

Q.3. Oral/Viva [5 Marks]

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-
- Q.1. Write the program to simulate Preemptive Shortest Job First (SJF) scheduling.
The arrival time and first CPU burst for different n number of processes should be input to the algorithm. The next CPU burst should be generated randomly. The output should give Gantt chart, turnaround time and waiting time for each process. Also find the average waiting time and turnaround time [15 Marks]
- Q.3 Write a C program to illustrate the concept of orphan process. Parent process creates a child and terminates before child has finished its task. So child process becomes orphan process. (Use fork(), sleep(), getpid(), getppid()) [15 Marks]
- Q.3. Oral/Viva [05 Marks]

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Q.1 Write the program to simulate Preemptive Priority scheduling. The arrival time and first CPU-burst and priority for different n number of processes should be input to the algorithm. The next CPU-burst should be generated. [15 Marks]

Q.2 Write a program that demonstrates the use of nice() system call.

After a child process is started using fork(), assign higher priority to the child using nice() system call [15 Marks]

Q.3. Oral/Viva [5 Marks]

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Q.1 Write a simulation program for disk scheduling using LOOK algorithm.

Accept total number of disk blocks, disk request string, and current head position from the user. Display the list of requests in the order in which it is served. Also display the total number of head moments.

86, 147, 91, 177, 45, 12, 130

Starting Head Position: 60

Direction: Right

[15 Marks]

Q.2 Write the simulation program to implement demand paging and show the page scheduling and total number of page faults according to the LRU (using counter method) page replacement algorithm. Assume the memory of n frames.

Reference String : 3,5,7,2,5,1,2,3,1,3,5,3,1,6,2

[15 Marks]

Q.3. Oral/Viva

[05 Marks]

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Q.1 Write the simulation program for demand paging and show the page scheduling and total number of page faults according the MFU page replacement algorithm. Assume the memory of n frames.

Reference String: 8, 5, 7, 8, 5, 7, 2, 3, 7, 3, 5, 9, 4, 6, 2 [15 Marks]

Q.2 Write a program that demonstrates the use of nice() system call.

After a child process is started using fork(), assign higher priority to the child using nice() system call. [15 Marks]

Q.3. Oral/Viva [05 Marks]

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Q.1 Write a program to simulate Linked file allocation method. Assume disk with 'n' number of blocks. Give value of 'n' as input. Randomly mark some block as allocated and accordingly maintain the list of free blocks Write menu driven program with menu options as mentioned below and implement each option.

- a. Show Bit Vector
- b. Create New File
- c. Show Directory
- d. Delete File
- e. Exit

[15 Marks]

Q.2 Write the program to simulate Non-preemptive Priority scheduling.

The arrival time and first CPU-burst for different n number of processes should be input to the algorithm. The next CPU-burst should be generated randomly. The output should give Gantt chart, turnaround time and waiting time for each process. Also find the average waiting time and turnaround time.

[15 Marks]

Q.3. Oral/Viva

[05 Marks]

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-
- Q.1 Implement the C program to accept n integers to be sorted. Main function creates child process using fork system call. Parent process sorts the integers using bubble sort and waits for child process using wait system call. Child process sorts the integers using insertion sort [15 Marks]
- Q.2 Write the program to simulate Non-preemptive Shortest Job First (SJF) scheduling. The arrival time and first CPU-burst for different n number of processes should be input to the algorithm. The next CPU-burst should be generated randomly. The output should give Gantt chart, turnaround time and waiting time for each process. Also find the average waiting time and turnaround time. [15 Marks]
- Q.3. Oral/Viva [5 Marks]

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-
- Q.1 Write the program to simulate Round Robin (RR) scheduling. The arrival time and first CPU-burst for different n number of processes should be input to the algorithm. Also give the time quantum as input. The next CPU burst should be generated randomly. The output should give Gantt chart, turnaround time and waiting time for each process. Also find the average waiting time and turnaround time. [15 Marks]
- Q.2 Implement the C program that accepts an integer array. Main function forks child process. Parent process sorts an integer array and passes the sorted array to child process through the command line arguments of `execve()` system call. The child process uses `execve()` system call to load new program that uses this sorted array for performing the binary search the particular item in the array. [15 Marks]
- Q.3. Oral/Viva [5 Marks]

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Q.1 Write a C program to illustrate the concept of orphan process. Parent process creates a child and terminates before child has finished its task. So child process becomes orphan process. (Use fork(), sleep(), getpid(), getppid())

[15 Marks]

Q.2 Write a C program that behaves like a shell which displays the command prompt '\$'. It accepts the command, tokenize the command line and execute it by creating the child process. Also implement the additional command 'search' as

- a. \$ search f filename pattern : It will search the first occurrence of pattern in the given file
- b. \$ search a filename pattern : It will search all the occurrence of pattern in the given file
- c. \$ search c filename pattern : It will count the number of occurrence of pattern in the given file

[15 Marks]

Q.3. Oral/Viva

[5 Marks]

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Q.1 Write the program to simulate Preemptive Priority scheduling. The arrival time and first CPU-burst and priority for different n number of processes should be input to the algorithm. The next CPU-burst should be generated

[15 Marks]

Q.2 Write the simulation program for demand paging and show the page scheduling and total number of page faults according the FIFO page replacement algorithm. Assume the memory of n frames.

Reference String: 3, 4, 5, 6, 3, 4, 7, 3, 4, 5, 6, 7, 2, 4, 6

[15 Marks]

Q.3. Oral/Viva

[05 Marks]

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Q.1 Write a simulation program for disk scheduling using SSTF algorithm.

Accept total number of disk blocks, disk request string, and current head position from the user. Display the list of requests in the order in which it is served. Also display the total number of head moments.

30, 10, 60, 95, 120, 150, 175

Start Head Position: 50

[15 Marks]

Q.2 Write the simulation program to implement demand paging

and show the page scheduling and total number of page faults according to the LRU (using counter method) page replacement algorithm. Assume the memory of n frames.

Reference String : 3,5,7,2,5,1,2,3,1,3,5,3,1,6,2

[15 Marks]

Q.3. Oral/Viva

[05 Marks]

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Q.1 Write a simulation program for disk scheduling using SCAN algorithm.

Accept total number of disk blocks, disk request string, and current head position from the user. Display the list of requests in the order in which it is served. Also display the total number of head moments.

10, 25, 75, 90, 130, 145, 180, 55

Starting Head position= 80

Direction: Left

[15 Marks]

Q.2 Write the simulation program for demand paging and show the page

scheduling and total number of page faults according the MFU page replacement algorithm. Assume the memory of n frames.

Reference String: 8, 5, 7, 8, 5, 7, 2, 3, 7, 3, 5, 9, 4, 6, 2

[15 Marks]

Q.3. Oral/Viva

[5 Marks]

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-
- Q.1 Implement the C program that accepts an integer array. Main function forks child process. Parent process sorts an integer array and passes the sorted array to child process through the command line arguments of `execve()` system call. The child process uses `execve()` system call to load new program that uses this sorted array for performing the binary search the particular item in the array [15 Marks]
- Q.2 Write the program to simulate FCFS CPU-scheduling. The arrival time and first CPU- burst for different n number of processes should be input to the algorithm. The next CPU-burst should be generated randomly. The output should give Gantt chart, turnaround time and waiting time for each process. Also find the average waiting time and turnaround time. [15 Marks]
- Q.3. Oral/Viva [5 Marks]

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Q.1 Write the program to simulate Round Robin (RR) CPU-scheduling. The arrival time and first CPU-burst for different n number of processes should be input to the algorithm. Also give the time quantum as input. Assume the fixed IO waiting time (2 units). The next CPU-burst should be generated randomly. The output should give Gantt chart, turnaround time and waiting time for each process. Also find the average waiting time and turnaround time

[15 Marks]

Q.2 Write the simulation program for demand paging and show the page scheduling and total number of page faults according to the MFU page replacement algorithm. Assume the memory of n frames.

Reference String: 2,5,2,8,5,4,1,2,3,2,6,1,2,5,9,8

[15 Marks]

Q. 3 Oral/Viva

[05 Marks]